Predicting VA Loan Approval Outcomes: A Machine Learning Classification and Fairness Analysis Using HMDA Data

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Abstract

The U.S. Department of Veterans Affairs home loan program was created to expand homeownership access for service members and veterans, offering zero down payment financing without private mortgage insurance. Despite this design, applications are still denied, and decades of research on conventional mortgage lending has documented approval disparities across demographic groups. This study examines what drives VA loan approval decisions and whether machine learning models trained on those decisions treat demographic groups differently. Using 339,211 VA-guaranteed home purchase applications from the 2025 Home Mortgage Disclosure Act (HMDA) national dataset, three classification models were compared: a logistic regression baseline, a random forest, and a gradient boosting classifier. Gradient boosting achieved the strongest discrimination (AUC = 0.799) against the linear baseline (AUC = 0.733), confirming that nonlinear structure exists in the approval decision. The most consequential finding concerns the shape of that structure. Published research on the broader mortgage market documents a sharp increase in denial rates at a 50 percent debt-to-income (DTI) ratio. VA lending does not follow this pattern. The denial rate rises only 0.7 percentage points at the 50 percent mark, but rises 28.5 percentage points above 60 percent, and the relationship is non-monotonic at the low end, with applicants below 20 percent DTI denied more often than those between 20 and 30 percent. A two-stage fairness analysis measured demographic parity difference, equalized odds, equal opportunity, and calibration across groups, then decomposed observed gaps using model predictions built only from underwriting variables. Underwriting factors accounted for 46.1 percent of the Black-White approval gap, 56.8 percent of the Hispanic gap, and 66.2 percent of the gap for female applicants, leaving unexplained residuals in each case. These residuals do not establish discrimination, since the public HMDA file omits credit score, the single most important underwriting variable. All findings were verified independently in JMP Pro, where debt-to-income ratio ranked as the strongest effect by a factor of more than three (logworth 2,817.96) and no parameter reversed sign relative to the Python estimates. The results indicate that VA underwriting operates on a different risk logic than conventional lending and that fairness auditing of VA lending models requires underwriting controls before any disparity can be interpreted.

Keywords: VA loans, mortgage lending, machine learning, algorithmic fairness, HMDA, veteran homeownership

1. Introduction

1.1 Research Problem

I work as a licensed real estate agent in Hawaii, and my clients are almost entirely active-duty military and veterans. I have watched qualified buyers get denied for VA loans and walk away believing homeownership is out of reach. The VA home loan program was created as an earned benefit, with zero down payment, no private mortgage insurance, and more forgiving underwriting, specifically to expand homeownership access for those who have served. Yet denials still happen at meaningful rates, and decades of mortgage research on conventional lending has documented persistent disparities in approval outcomes across demographic groups.

That is the tension driving this study. A loan program explicitly designed to open doors may still produce outcomes that close them for some applicants. If the program works as intended, approval decisions should be driven by financial factors such as debt-to-income ratio and loan amount. If disparities exist in VA lending the way they do in conventional lending, that contradicts the purpose of the benefit. Both possibilities are worth testing rather than assuming.

There is a second, methodological layer to the problem. Mortgage underwriting is increasingly automated, and machine learning models trained on historical lending decisions will learn whatever patterns exist in those decisions, including unfair ones. Before the industry leans further into algorithmic underwriting for VA loans, it is worth checking what these models actually learn and whether they treat demographic groups differently.

Problem statement. The VA loan program is designed to expand homeownership access for service members and veterans, yet the determinants of VA loan approval outcomes, and the fairness properties of machine learning models trained to predict those outcomes, have not been systematically examined using large-scale public lending data.

1.2 Research Questions

RQ1. Which applicant, loan, and property characteristics most strongly predict VA loan approval outcomes in HMDA data?

RQ2. Do machine learning models that capture nonlinear relationships outperform a logistic regression baseline in predicting VA loan approval, as measured by discrimination and calibration metrics, and if so, which nonlinear patterns explain the difference?

RQ3. Do the trained models exhibit disparate error rates or outcome rates across demographic groups, as measured by demographic parity difference, equalized odds, equal opportunity, and calibration across groups, and to what extent are observed disparities attributable to differences in underwriting factors versus left unexplained after controls?

RQ4. Do the drivers of VA loan approval in the Hawaii market differ from national patterns?

2. Literature Review

2.1 Mortgage Approval Disparities

The empirical literature on mortgage lending disparities begins with Munnell, Tootell, Browne, and McEneaney (1996), who matched HMDA records to applicant credit files for Boston-area lenders and found that Black and Hispanic applicants remained substantially more likely to be denied than White applicants after controlling for financial, employment, and neighborhood characteristics. That study established both the central empirical claim and the methodological template that subsequent work has followed.

Recent research using the post-2018 expanded HMDA fields has complicated the picture. Popick (2022) found that credit-factor and lender controls explain roughly 70 percent of the conventional denial-rate gap, with meaningful disparities remaining. Bhutta, Hizmo, and Ringo (2024) reported that raw Black-White denial gaps of approximately 12 percentage points shrink to 1 to 2 percentage points of excess denials once observable underwriting factors are controlled, and argued that even that residual may reflect unobserved factors. They also noted that group differences in the underwriting factors themselves partly trace to broader inequities, which means the decomposition answers a narrower question than it first appears to.

This debate is directly relevant to the present study. It establishes that raw approval gaps are not self-interpreting, and that any credible fairness analysis must separate the measurement of a disparity from the explanation of it.

2.2 VA Loans and Veteran Homeownership

Fetter (2013) provides the canonical causal evidence on VA loan benefits, using a regression discontinuity around draft-eligibility cutoffs to show that mid-century GI Bill loan guaranty provisions raised homeownership primarily by shifting purchase earlier in life. This establishes the program's historical effectiveness and its purpose as an access-expanding intervention.

Research specifically examining fairness in VA lending is sparse. Work on the geography of VA loan usage suggests utilization concentrates near major military installations and is lower than expected in high-cost markets, which motivates the Hawaii subset analysis in RQ4. The relative absence of veteran-focused fairness research is the gap this study addresses, though following the guidance that a gap alone is not a rationale, the primary justification here remains the substantive contradiction described in Section 1.1.

2.3 Algorithmic Fairness in Lending

Bartlett, Morse, Stanton, and Wallace (2022) provide the central finding for this study's third research question. Examining both algorithmic and face-to-face lending, they found that minority borrowers are charged higher rates on purchase and refinance mortgages, and that FinTech algorithms also discriminate, though roughly 40 percent less than human loan officers. Algorithmic lending therefore neither eliminates nor is immune to disparate treatment.

On fairness measurement, Hardt, Price, and Srebro (2016) introduced equalized odds and equality of opportunity as criteria that condition on the true outcome rather than on raw selection rates, which matters when groups differ in their underlying distribution of underwriting characteristics. Mehrabi, Morstatter, Saxena, Lerman, and Galstyan (2021) survey the broader taxonomy of bias sources and fairness definitions. Lee and Floridi (2021) applied multiple fairness definitions to HMDA data directly and demonstrated that these definitions conflict, so that no single model satisfies all of them simultaneously. Their conclusion, that fairness in lending involves relational trade-offs rather than absolute conditions, shapes how the present study reports its results: as a set of complementary measurements rather than a single verdict.

3. Method

3.1 Data Source

This study uses archival loan-level records from the Home Mortgage Disclosure Act (HMDA) database, downloaded from the HMDA Data Browser maintained by the Consumer Financial Protection Bureau (CFPB) and the Federal Financial Institutions Examination Council (FFIEC) at ffiec.cfpb.gov. HMDA data is not voluntary or self-reported by borrowers. It is collected under federal statute, the Home Mortgage Disclosure Act of 1975, implemented through Regulation C, and most mortgage lenders in the United States are legally required to report application-level records annually. Since 2018 the public files include expanded fields covering debt-to-income ratio, loan-to-value ratio, loan costs, and detailed demographic identifiers.

The extract covers filing year 2025, filtered at download to VA-guaranteed first-lien home purchase applications on single-family site-built properties, yielding 617,940 raw records across 99 fields. Because the data is public and de-identified at the application level, no IRB approval was required.

3.2 Data Screening

Screening proceeded in documented steps, with record counts retained at each stage so that every filtering decision is transparent and reproducible. Table 1 reports the full sequence.

Screening step	**Before**	**After**	**Removed**
Raw download (VA, home purchase, 2025)	617,940	617,940	0
First lien only	617,940	617,903	37
Principal residence only	617,903	616,862	1,041
Site-built only	616,862	599,250	17,612
Single family only	599,250	592,354	6,896
Exclude reverse mortgages	592,354	591,287	1,067
Exclude business purpose	591,287	591,182	105
Exclude preapproval requests	591,182	569,964	21,218
Lender decision only	569,964	340,692	229,272
Income plausibility	340,692	339,255	1,437
Loan amount plausibility	339,255	339,244	11

Screening step	Before	After	Removed
LTV plausibility	339,244	339,211	33

Two screening decisions warrant explanation. First, the outcome variable retains only records reflecting an actual lender decision on creditworthiness: originated loans (action taken code 1) and applications approved but not accepted (code 2) were coded as approved, and denials (code 3) as denied. Withdrawn applications, files closed for incompleteness, and purchased loans were excluded because they do not represent a credit decision. This removed 229,272 records, the single largest exclusion.

Second, applications involving a preapproval request were excluded entirely. When a preapproval request is denied, HMDA assigns action taken code 7, which falls outside the outcome definition above. Retaining preapproval records would therefore have left an approval-only subgroup of 11,927 records containing zero denials, producing perfect separation on the preapproval field. Excluding these applications removes that artifact at the cost of 21,218 records.

The final analytic sample is 339,211 applications with an overall approval rate of 91.84 percent.

3.3 Target Leakage Controls

A substantial share of HMDA fields describe loan terms that exist only once a loan has been made. Including any of them would produce a model that appears highly accurate while learning nothing about the approval decision. Fourteen fields were therefore excluded, in two categories.

Six fields are structurally missing for all denied applications: interest rate, rate spread, total loan costs, total points and fees, origination charges, discount points, and lender credits. A model given these fields can identify denials simply by detecting missingness.

Two further fields are fully populated but encode the outcome through their category codes. HOEPA status carries codes 1 and 2 for loans with a determinable high-cost status, both of which show 100 percent approval, and code 3, not applicable, which shows 23.6 percent approval; its univariate AUC against the outcome is 0.987. The field indicating whether the loan was initially payable to the reporting institution shows the same structure, with its not-applicable

code at 23.9 percent approval. Denial reason and purchaser type were excluded on the same grounds.

One borderline field was retained. The automated underwriting system indicator shows 93.2 percent approval when Desktop Underwriter was used and 65.3 percent when no automated system was used, indicating manual underwriting. This is a genuine pre-decision process characteristic rather than a post-decision artifact, and it was retained with this caveat noted.

3.4 Features and Models

Nine numeric features were used: loan amount, income, property value, loan term, loan-to-value ratio, a numeric DTI lower bound, tract minority population percentage, tract-to-MSA income percentage, and FFIEC MSA median family income. Eight categorical features were used: DTI bin, applicant and co-applicant credit score type, automated underwriting system, conforming loan limit status, applicant age band, submission channel, and state.

Debt-to-income ratio appears in the public file as a mixture of exact integer values between 36 and 49 and binned ranges outside that window. It was therefore treated as an ordered categorical variable with a parallel numeric lower-bound encoding, which permits direct comparison of adjacent bins around any candidate threshold.

Protected attributes (derived race, derived ethnicity, derived sex) were deliberately excluded from the feature matrix and used only for fairness evaluation. Demographic fields were never imputed, since imputing race or sex would be inappropriate in a fairness analysis; records with unavailable demographic data were retained for modeling but excluded from group-level fairness comparisons.

Three models were fit on a stratified 75/25 train-test split: a logistic regression baseline with balanced class weights, a random forest (250 trees, minimum leaf size 25, balanced class weights), and a histogram-based gradient boosting classifier (300 iterations, learning rate 0.08). Numeric features were median-imputed and standardized; categorical features were one-hot encoded with a minimum category frequency of 100.

3.5 Evaluation and Fairness Metrics

Model discrimination was assessed by AUC-ROC and F1, and probability quality by Brier score and calibration curves. Because approvals outnumber denials by more than ten to one, accuracy alone would be misleading, and a 0.50 decision threshold approves 97.0 percent of applicants, which makes false positive rates uninformative. Fairness metrics were therefore computed at the Youden-optimal threshold (0.9253), which approves 81.6 percent of applicants and yields interpretable error rates in both directions.

Four fairness measures were computed, following the two-stage design developed in response to mentor feedback that statistical disparity does not by itself establish unfairness. Stage one measures: demographic parity difference as a descriptive baseline only, equalized odds and equal opportunity (Hardt et al., 2016) as primary metrics because they condition on the true outcome, and calibration error across groups. Stage two decomposes each observed gap by comparing the raw difference in approval rates against the difference the model predicts from underwriting variables alone. Because the model has no access to demographic attributes, its predicted gap represents the portion attributable to differences in financial characteristics, and the remainder is the unexplained residual.

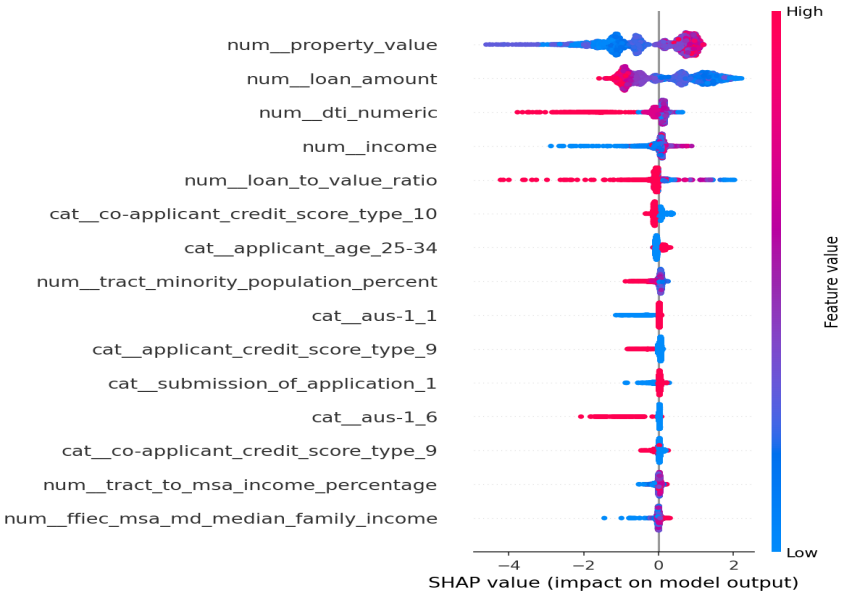
Analysis was conducted in Python 3.11 using scikit-learn, SHAP, pandas, and matplotlib. The logistic baseline and the debt-to-income analysis were then rebuilt independently in JMP Pro from an identical exported dataset, so that any discrepancy between environments would reflect method rather than data. Appendix A reports that comparison in full.

4. Results

4.1 RQ1: Predictors of VA Loan Approval

SHAP attribution on the best-performing model ranks property value (mean absolute SHAP = 0.976) and loan amount (0.892) as the dominant predictors by a wide margin, followed by debt-to-income ratio (0.220), income (0.132), and loan-to-value ratio (0.119). Neighborhood context variables contribute far less: tract minority population percentage registers 0.077 and tract-to-MSA income percentage 0.042.

The prominence of property value and loan amount relative to DTI is notable. It suggests that in VA purchase lending, the size and valuation of the transaction carry more predictive weight than the applicant's ratio-based risk profile, at least among the variables the public file discloses.

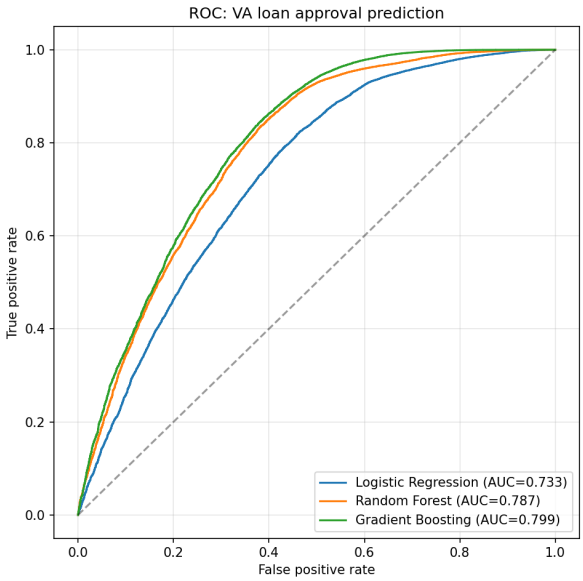


4.2 RQ2: Model Comparison and Nonlinearity

All three models were evaluated on the same held-out test set of 84,803 applications. Results appear in Table 2.

Model	**AUC-ROC**	**F1**	**Brier**	**Accuracy**
Logistic regression (baseline)	0.7328	0.8826	0.1919	0.7990
Random forest	0.7858	0.9292	0.1393	0.8737
Gradient boosting	0.7995	0.9668	0.0539	0.9374

Gradient boosting improves AUC by 0.0667 over the linear baseline and reduces the Brier score by more than two thirds, from 0.1919 to 0.0539. The answer to the first half of RQ2 is therefore affirmative: nonlinear structure exists in the VA approval decision and flexible models capture it.

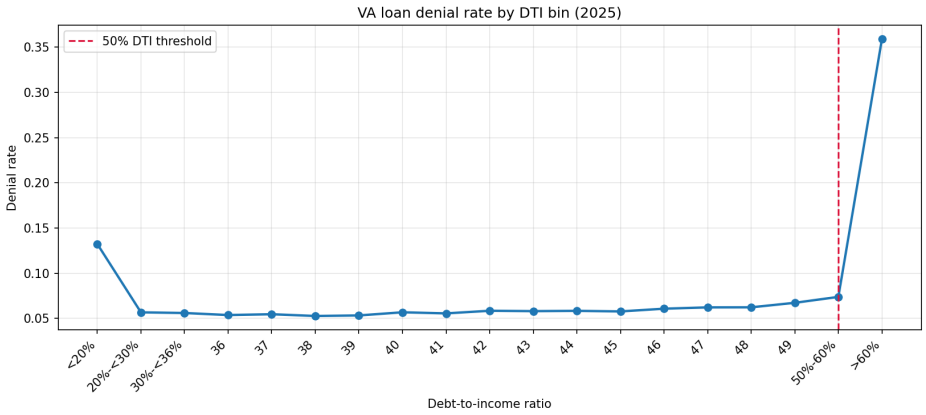


The second half of RQ2 asks which nonlinear patterns explain the difference, and the answer produced the study's most substantive finding. Published analysis of the broader mortgage market documents a sharp increase in denial rates at a 50 percent debt-to-income ratio, reported at 15 to 17 percentage points. VA lending does not reproduce this pattern. Table 3 reports denial rates across the full DTI distribution.

DTI bin	**n**	**Denial rate**
Below 20%	6,036	13.22%
20% to under 30%	30,689	5.66%
30% to under 36%	40,203	5.59%
36%	8,279	5.36%
40%	10,531	5.67%
45%	11,955	5.76%
49%	12,905	6.72%
50% to 60%	86,401	7.37%
Above 60%	21,296	35.91%

Two departures from the conventional pattern are visible. First, crossing the 50 percent threshold raises the denial rate by only 0.7 percentage points, from 6.72 to 7.37 percent. The cliff documented for the general mortgage market is essentially absent in VA lending. Second, a far larger discontinuity appears above 60 percent, where the denial rate jumps 28.5 percentage points, from 7.37 to 35.91 percent.

The relationship is also non-monotonic at the low end. Applicants below 20 percent DTI are denied at 13.22 percent, more than double the 5.66 percent rate for applicants between 20 and 30 percent. Low debt-to-income is therefore not simply protective, and a linear specification would not merely lose precision here, it would invert the relationship over part of its range.



This pattern is consistent with VA underwriting practice, which applies a residual income methodology rather than a hard DTI ceiling. VA guidelines flag ratios above 41 percent for additional scrutiny but permit substantially higher ratios when residual income is adequate. The binding constraint therefore appears where the conventional market has already stopped lending.

4.3 RQ3: Fairness Analysis

Stage one measured the four fairness metrics across the three protected attributes at the Youden-optimal threshold. Results appear in Table 4. Groups with fewer than 200 test records, and records with unavailable demographic data, were excluded from these comparisons.

Attribute	Demographic parity diff.	Equal opportunity diff.	Equalized odds diff.	Max calibration error
Race	0.1133	0.0927	0.0927	0.0383
Ethnicity	0.0685	0.0614	0.0614	0.0071
Sex	0.0858	0.0699	0.0699	0.0049

Demographic parity differences are consistently larger than equal opportunity differences across all three attributes, which is the expected signature when groups differ in their underlying distribution of underwriting characteristics. Calibration is strong for ethnicity and sex, with maximum errors below 0.008, and weaker for race at 0.0383.

Stage two decomposes the observed approval gaps. Because the model receives no demographic information, its predicted gap between two groups reflects only differences in their underwriting profiles. Table 5 reports the decomposition against the largest group in each attribute.

Group (vs. reference)	n	Raw gap	Explained	Residual	% explained
Black or African American (vs. White)	10,397	-6.67 pp	-3.08 pp	-3.59 pp	46.1%
2 or more minority races (vs. White)	401	-6.77 pp	-2.34 pp	-4.43 pp	34.6%
American Indian / Alaska Native (vs. White)	854	-3.47 pp	-2.44 pp	-1.03 pp	70.2%
Native Hawaiian / Pacific Islander (vs. White)	306	-1.87 pp	-1.11 pp	-0.77 pp	59.2%
Asian (vs. White)	2,342	-0.48 pp	-0.45 pp	-0.03 pp	94.1%
Hispanic or Latino (vs. Not Hispanic)	9,531	-1.88 pp	-1.07 pp	-0.81 pp	56.8%
Female (vs. Male)	8,859	-1.35 pp	-0.90 pp	-0.46 pp	66.2%

The largest raw gap is for Black applicants, at 6.67 percentage points. Underwriting variables account for 46.1 percent of it, leaving an unexplained residual of 3.59 percentage points. For Asian applicants the gap is both small and almost entirely explained. For applicants reporting two or more minority races the explained share is lowest at 34.6 percent, though the small sample of 401 test records warrants caution.

These residuals are reported as measurements, not as findings of discrimination. Section 5.2 addresses their interpretation directly.

4.4 RQ4: The Hawaii Market

Applying the same pipeline to Hawaii applications yields 2,217 records after screening, with 161 denials. The Hawaii approval rate of 92.74 percent is slightly above the national 91.84 percent. Model performance is lower, with the gradient boosting AUC at 0.7363 against a logistic baseline of 0.6775, a gap of 0.0587 that closely matches the national improvement of 0.0667.

The DTI pattern replicates in direction: the denial rate above 60 percent reaches 24.06 percent against 5.96 percent in the 50 to 60 percent band, an 18.1 percentage point discontinuity. Because the Hawaii sample contains only 161 denials distributed across nineteen DTI bins and several demographic groups fall below the 200-record threshold, group-level fairness metrics could not be estimated reliably. The Hawaii analysis is therefore reported as descriptive corroboration of the national DTI finding rather than as an independent fairness audit.

5. Discussion

5.1 VA Underwriting Follows a Different Risk Logic

The central empirical contribution of this study is that the DTI threshold documented for the broader mortgage market does not transfer to VA lending. The conventional market's 50 percent cliff is nearly flat here, and the operative discontinuity sits above 60 percent. Combined with the non-monotonic elevation in denial rates among the lowest-DTI applicants, this indicates that VA underwriting is not a more permissive version of conventional underwriting but a structurally different decision process.

The practical implication is direct. Guidance that treats 50 percent DTI as the decisive line, whether given by agents, lenders, or automated tools calibrated on conventional data, is miscalibrated for VA applicants. The evidence here suggests VA borrowers retain meaningful approval odds well past that point and encounter a genuine constraint closer to 60 percent.

The elevated denial rate below 20 percent DTI deserves further study. It may reflect applicants with low reported income relative to obligations, unusual file characteristics, or compositional differences not captured in the public fields. It is a finding this study surfaces rather than explains.

5.2 Interpreting the Fairness Results

The decomposition in Table 5 shows unexplained residuals for most demographic groups, with the largest at 3.59 percentage points for Black applicants. Three considerations bound what can be concluded from this.

First, and most importantly, the public HMDA file does not contain credit score. Credit score is the single most influential variable in mortgage underwriting, and its absence means the underwriting controls in this analysis are incomplete by construction. Bhutta, Hizmo, and Ringo (2024), working with confidential data that includes credit scores, found that raw gaps of roughly 12 percentage points shrink to 1 to 2 points once full underwriting information is available. The residuals reported here are therefore upper bounds on unexplained disparity, not estimates of discrimination. This is also why the models plateau near an AUC of 0.80: the most predictive variable is simply unavailable.

Second, as noted in the same literature, differences in the underwriting factors themselves are not exogenous. A gap fully explained by DTI and loan-to-value has been attributed to financial characteristics, but those characteristics have their own histories. The decomposition answers a narrower question than the phrase explained gap suggests.

Third, the fairness metrics disagree with one another in the manner Lee and Floridi (2021) predict. Demographic parity differences exceed equal opportunity differences for every attribute, and calibration is strong for sex and ethnicity while weaker for race. No single number summarizes fairness here, and selecting one metric would amount to choosing a conclusion.

What the analysis does establish is narrower and still useful: a measurable portion of the observed approval gap in VA lending is not accounted for by the financial variables that are publicly observable, and that portion is largest for the groups with the largest raw gaps. This is a finding that justifies further examination with richer data, not a verdict.

5.3 Limitations

Two limitations surfaced only through the cross-tool comparison in Appendix A. First, the ordered categorical debt-to-income variable and its numeric encoding are deterministically related. The L2 penalty in the Python implementation absorbed that collinearity silently, while the unpenalized JMP fit exposed it as aliased parameters and standard errors exceeding 12,000. The numeric encoding was removed from the JMP specification, and the episode is a reminder that regularization can conceal specification problems rather than resolve them. Second, the two environments handle missing data differently: Python median-imputed and retained 339,211 records, while JMP applied listwise deletion and fit 268,569. The convergence of results across a 70,642-record difference is reassuring, but it also means the estimated influence of loan-to-value ratio is attenuated in the Python model.

Missingness itself appears informative rather than random. Applications with no reported debt-to-income ratio are denied at approximately 17 percent, roughly three times the rate of the modal bins. Median imputation therefore assumes something the data does not support, and a sensitivity analysis using a missing-indicator specification would be a worthwhile extension.

One further data-quality artifact deserves note. Records with a blank state field carry the largest coefficient in the logistic model, with an odds ratio near 0.08 in both environments. This reflects incomplete geographic reporting rather than a substantive finding about location, and it is excluded from the interpretation in Section 4.1.

Beyond the omission of credit score, four limitations apply. The analysis covers a single filing year, so results may reflect 2025 market conditions rather than stable structure. Roughly 20 percent of records lack loan-to-value ratio and were median-imputed, which attenuates that variable's estimated influence. The retained automated underwriting system indicator sits close to the boundary between process characteristic and outcome proxy, and excluding it would modestly reduce measured performance. Finally, the Hawaii subset is too small to support group-level fairness inference, and its findings are descriptive only.

6. Conclusion and Practical Implications

This study analyzed 339,211 VA home purchase applications to identify the drivers of approval, test whether nonlinear models capture structure that linear models miss, and audit the fairness properties of the resulting predictions.

Three conclusions follow. First, nonlinear models meaningfully outperform a linear baseline, with gradient boosting improving AUC from 0.733 to 0.799 and reducing Brier score by more than two thirds. Second, and more consequentially, the nonlinearity takes a shape specific to VA lending: the 50 percent DTI cliff documented for conventional mortgages is nearly absent, while a 28.5 percentage point discontinuity appears above 60 percent, and the relationship reverses direction below 20 percent. Third, underwriting variables explain between 34.6 and 94.1 percent of demographic approval gaps depending on the group, leaving residuals that are real but not interpretable as discrimination given the absence of credit score from the public file.

For veterans and the professionals who serve them, the DTI finding is immediately actionable. A VA applicant at 52 percent DTI faces a denial rate of 7.37 percent, close to that of an applicant at 45 percent, and guidance calibrated to conventional lending thresholds understates their prospects. The constraint that matters sits nearer 60 percent.

For lenders and regulators, the results argue that fairness auditing of VA lending models requires underwriting controls before any disparity is interpreted, and that models calibrated on conventional lending data should not be assumed to transfer to VA underwriting, whose risk logic differs materially.

For future research, the natural extension is to replicate this analysis with credit score data, which would convert the upper-bound residuals reported here into interpretable estimates. Multi-year replication would establish whether the 60 percent discontinuity is stable, and a larger geographic sample would permit the market-level fairness comparison that the Hawaii subset could not support.

Appendix A: Software Triangulation

A.1 Approach

Consistent with the multi-software emphasis of this program, the analysis was conducted independently in two environments. Python 3.11 with scikit-learn and SHAP served as the primary modeling environment. JMP Pro was used as an independent check on the logistic regression baseline and on the principal debt-to-income finding.

Both tools operated on an identical dataset. The screened analytic sample was exported from the Python pipeline as a single file containing all 339,211 records and the 22 relevant columns, and imported into JMP without further modification. Any disagreement therefore reflects differences in method rather than in the underlying data.

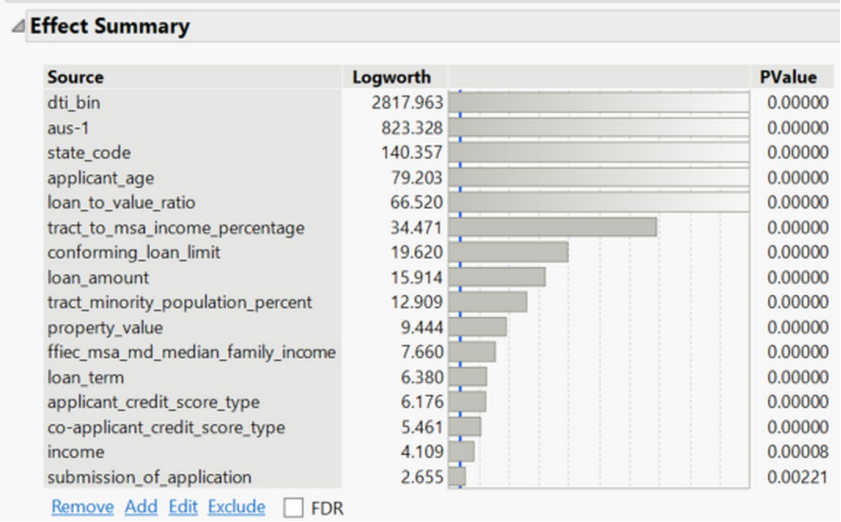
Three implementation differences should be understood before comparing results. First, scikit-learn applies one-hot encoding with a dropped reference level and L2 regularization by default, while JMP uses effect coding and unpenalized maximum likelihood, so coefficient magnitudes are not directly comparable and the appropriate comparison is direction, ranking, and rough magnitude. Second, the Python baseline applied balanced class weights to address the roughly ten to one outcome imbalance, while JMP fit the model unweighted. Third, Python median-imputed missing numeric values while JMP applied listwise deletion, fitting 268,569 observations against Python's 339,211.

A.2 Specification Adjustment

An initial JMP run including both the ordered categorical debt-to-income variable and its numeric lower-bound encoding produced aliased parameters, lost degrees of freedom on the categorical term, and standard errors exceeding 12,000 on several effects. The two variables are deterministically related, and the unpenalized fit cannot accommodate that collinearity, whereas the L2 penalty in Python absorbs it without visible instability. The numeric encoding was removed. After removal all sixteen effects were significant, no degrees of freedom were lost, and the estimates stabilized.

A.3 Convergent Results

JMP ranks effects by logworth, the negative base-ten logarithm of the p-value. This is a different basis than the SHAP attribution used in Python, which measures average contribution to individual predictions rather than statistical significance. The two rankings nevertheless converge on the same substantive conclusion.



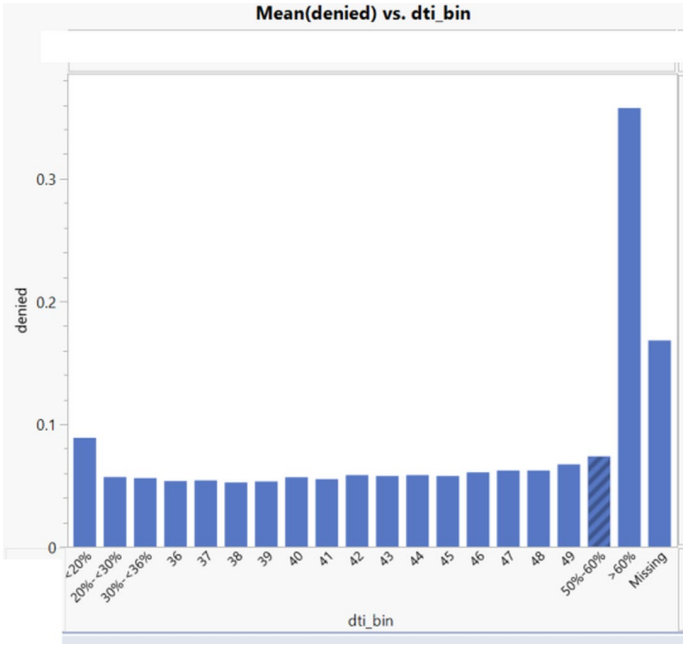
Debt-to-income ratio ranks first at a logworth of 2,817.96, more than three times the next effect. In the Python analysis the same variable was the source of the nonlinear structure separating the tree-based models from the linear baseline. Two methodologically distinct approaches converge on debt-to-income as the dominant driver of VA approval outcomes.

Effect	JMP odds ratio	Python odds ratio	Agreement
state_code (blank)	0.084	0.081	Close
state_code (PR)	0.360	0.392	Close
state_code (HI)	1.440	1.492	Close
dti_bin above 60%	0.099	0.176	Same direction, strong effect
dti_bin below 20%	0.844	0.448	Same direction
loan_to_value_ratio	Negative, chi-square 298	Negative	Same direction
aus-1 level 6 (manual)	Lowest of aus-1 levels	Lowest of aus-1 levels	Same ranking

No effect reversed sign between environments. The geographic estimates, which are unaffected by the weighting and regularization differences, agree to within approximately four percent. The debt-to-income and automated underwriting effects agree in direction and in relative ordering within each variable.

A.4 Independent Reproduction of the Principal Finding

The debt-to-income distribution was rebuilt in JMP Graph Builder as the mean of a denial indicator across bins, with bin ordering set explicitly rather than alphabetically.



The JMP rendering reproduces the Python result: an elevated denial rate below 20 percent, a flat plateau near 5.5 percent across the middle of the distribution, a modest step at the 50 to 60 percent band, and a pronounced discontinuity above 60 percent reaching approximately 36 percent. Because this is a direct calculation on identical data rather than a model estimate, exact agreement confirms correct data transfer. Its value lies in visual confirmation that the shape of the relationship, including the non-monotonic elevation at the low end, is a property of the data rather than an artifact of the Python pipeline.

The chart also displays the Missing category at approximately 17 percent, which is the basis for the observation in Section 5.3 that missing debt-to-income is itself informative.

A.5 Conclusion

The two environments agree on every substantive conclusion of this study. Debt-to-income ratio is the dominant predictor in both. The shape of that relationship, including the absence of a threshold effect at 50 percent and the sharp discontinuity above 60 percent, is identical in both. No effect reverses sign, and directly comparable parameters agree closely.

The exercise also produced two findings a single-tool analysis would not have surfaced: the collinearity between the two debt-to-income encodings, invisible under regularization and immediately apparent without it, and the sensitivity of the analytic sample to missing-data handling, a difference of 70,642 records. Both are reported in the limitations rather than resolved, and both improve the transparency of the analysis.

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